Inverse z-Transforms via Power Series Expansion

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Key Concepts

1) The inverse z-transform finds the time signal from the z-transform $X(z)$.

2) Inversion using a power series expansion writes $X(z)$ in the form of a power series.

3) The value $x[n]$ is the coefficient of z^{-n} in the power series expression of the z-transform.

4) Power series expansion is particularly useful for inverting transcendental functions of z.

5) Rational z-transforms can be inverted by power series expansion by using long division.

Video Screenshots

0:10

z-Transform is a power series

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

$$= \ldots + x[-2]z^{2} + x[-1]z + x[0] + x[1]z^{-1} + x[2]z^{-2} + \ldots$$

1) Write function to be inverted as a power series
2) Identify $x[n]$ as coefficient of z^{-n}

Example:

$$X(z) = 2z^{5} + z^{3} - z^{2} + 1 + 3z^{-1} - 4z^{-4}$$

$x[-5]$, $x[-3]$, $x[-2]$, $x[0]$, $x[1]$, $x[4]$

$$x[n] = 2\delta[n+5] + \delta[n+3] - \delta[n+2] + \delta[n] + 3\delta[n-1] - 4\delta[n-4]$$

Power series expansion can invert transcendental functions of z

Ex. $X(z) = \exp\{-2z^{-1}\}$

Recall $\exp\{x\} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ so

$$X(z) = \sum_{n=0}^{\infty} \frac{(-2z^{-1})^n}{n!} = \sum_{n=0}^{\infty} \frac{(-2)^n}{n!} z^{-n}$$

$$x[n] = \frac{(-2)^n}{n!} u[n]$$

Can also invert rational $X(z)$ with long division

Ex. $X(z) = \dfrac{1 - z^{-1}}{1 - \frac{1}{2} z^{-1}}$

$|z| > \frac{1}{2}$

$$1 - \tfrac{1}{2} z^{-1} \;\overline{)\; 1 - z^{-1}}$$

$$\begin{array}{r} 1 - \frac{1}{2} z^{-1} - \frac{1}{4} z^{-2} - \frac{1}{8} z^{-3} - \frac{1}{16} z^{-4} \cdots \\ \hline 1 - z^{-1} \\ 1 - \frac{1}{2} z^{-1} \\ \hline -\frac{1}{2} z^{-1} \\ -\frac{1}{2} z^{-1} + \frac{1}{4} z^{-2} \\ \hline -\frac{1}{4} z^{-2} \\ -\frac{1}{4} z^{-2} + \frac{1}{8} z^{-3} \\ \hline -\frac{1}{8} z^{-3} \\ -\frac{1}{8} z^{-3} + \frac{1}{16} z^{-4} \\ \hline -\frac{1}{16} z^{-4} \end{array}$$

$$x[n] = \begin{cases} 0 & n < 0 \\ 1 & n = 0 \\ -\frac{1}{2} & n = 1 \\ -\frac{1}{4} & n = 2 \\ -\frac{1}{8} & n = 3 \\ -\frac{1}{16} & n = 4 \\ \vdots & \end{cases}$$

$$x[n] = 2\,\delta[n] - \left(\tfrac{1}{2}\right)^n u[n]$$

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